Jay Kumar

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Professional Summary

Motivated Software Engineering student with strong command over Java and Python, specializing in Data Structures and Algorithms. Proficient in building intuitive iOS applications and driven by a passion for developing tech solutions that address real-life challenges. Working in integrating machine learning models into mobile apps and optimizing performance for practical deployment.

Skills

Programming: - Java, Python, C, C++ Javascript,React.

App Development: - Flutter (learning).

Data Analysis: - Pandas, NumPy, Matplotlib

Tools: - Git, GitHub, VS Code.

Concepts: - DSA, OOP, OS, Networks, DBMS, Machine Learning.

Soft Skills: - Team Building, Active Listening, Leadership.

Internship

GIS, Machine Learning & Deep Learning Internship – IIT Kharagpur *May 11 – Jul 11, 2026*

- Worked on integrating Geographic Information System (GIS) data with machine learning and deep learning models for spatial analysis and prediction.
- Preprocessed and analyzed satellite and remote sensing imagery using Python, GDAL, and QGIS for feature extraction.
- Built and trained CNN-based deep learning models for land-use and land-cover classification, improving prediction accuracy.
- Collaborated with research mentors to document methodology and optimize model performance on real-world geospatial datasets.

Full Stack Development & Java Programing Bootcamp *May 2025 – Jun 2025*

- Learned backend module of Java and develop project using Javascript and Nodejs
- Analyzed customer data using Pandas, NumPy, and Matplotlib..
- Collaborated via GitHub, applied OOP/DSA concepts.

Projects

AI – detection skin injurie machine learning model *Jan, 2025 Present*

- Developing an AI-based mobile application to detect and classify skin lesions using the HAM10000 dataset and EfficientNetB0 architecture.
- Integrated explainable AI techniques to enhance model transparency and user trust, with a clean UI for realtime predictions.
- Challenges Facing: Balancing model accuracy with lightweight deployment on mobile devices, and ensuring reliable classification across diverse skin tones and lesion types.

Travel Ginnie – A special Tour Guide for India *Dec, 2024*

- Developed a smart travel assistant app using Python, NLP, and machine learning to provide personalized itinerary suggestions, budget planning, and real-time travel updates.
- Designed to simplify trip planning for solo and group travelers by offering curated recommendations based on interests, location, and spending preferences.
- Challenges Faced: Ensuring accurate location-based suggestions with limited data, handling dynamic user queries, and optimizing response time across varied network conditions.

Banking Ledger System *2026*

- Designed and developed a full-fledged backend for a Banking Ledger System using Node.js, Express.js, and MongoDB.
- Implemented secure RESTful APIs for account management, fund transfers, and transaction history with JWT-based authentication.
- Applied double-entry bookkeeping logic to maintain ledger accuracy and prevent inconsistencies during concurrent transactions.
- Optimized database schema and queries, and added transaction rollback handling to ensure reliability of financial operations.

Education

B. Tech (Computer Science Engineering), Sharda University 2023-Present

AISSE (Class XII), CBSE – St. Karen's High School 2023